

Ain Shams University

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Resala Charity Management System

Software Requirements Specification

Course: CSE336 — Software Design Patterns

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Contents

1 Introduction **1**
1.1 Purpose . . . 1
1.2 Scope . . . 1
1.3 Definitions . . . 1
1.4 References . . . 1

2 Overall description **1**
2.1 Product perspective . . . 1
2.2 User classes . . . 2
2.3 Operating environment . . . 2
2.4 Constraints and assumptions . . . 2

3 Functional requirements **2**
3.1 Donations and donors . . . 2
3.2 Volunteers . . . 3
3.3 Events and attendance . . . 3
3.4 Beneficiaries and distribution . . . 3
3.5 Communication . . . 3
3.6 Reference data (lookup tables) . . . 4
3.7 Reports and documents . . . 4

4 Non-functional requirements **4**

5 External interfaces and data **4**
5.1 User interfaces . . . 4
5.2 Hardware and software interfaces . . . 5
5.3 Communication interfaces . . . 5
5.4 Data requirements . . . 5

6 Traceability and scope limits **5**
6.1 Client requirements to functional requirements . . . 5
6.2 Out of scope for this version . . . 6
6.3 Acceptance of the phase . . . 6

1 Introduction

1.1 Purpose

This document specifies the requirements for the Resala Charity Management System, a desktop application built for a non-profit organization. It is written for the course instructor and for the development team, and it defines what the system must do before the design and implementation are judged. The document covers the Phase 1 scope and states the limits of that scope explicitly.

1.2 Scope

The system supports the daily operations of a charity branch: donations and donors, volunteers and their skills and hours, events and attendance, beneficiaries and aid distribution, communication with stakeholders, and the reference data used by these modules. It runs on one computer, stores its data locally, and offers both a graphical interface and a console interface.

The following items are inside the scope of this version:

- recording and managing donors, donations, volunteers, events, beneficiaries, and distributions;
- generating receipts, yearly statements, attendance reports, and volunteer certificates;
- validating volunteer assignment against availability, skill, and event capacity;
- notifying interested parties when a donation is recorded or an event changes;
- selecting a communication channel at runtime;
- maintaining event types and skills as editable reference data.

1.3 Definitions

Term	Meaning
Donor	A person or organization that gives money or goods to the charity.
Donation	A recorded contribution. It can be online, by check, or in-kind.
In-kind donation	A donation of goods rather than money, described by an item and a quantity.
Volunteer	A person who offers time and skills to the charity.
Skill	A capability that can be attached to a volunteer and required by an event.
Event	An organized activity such as a fundraiser, workshop, or outreach program.
Registration	The record that a person will attend an event.
Beneficiary	A person or family that receives aid from the charity.
Distribution	A recorded delivery of aid to a beneficiary.
Lookup table	Editable reference data used by other modules, such as event types and skills.
Receipt	A document produced after a donation, optionally with tax and thank-you sections.
Statement	A yearly summary of a donor's monetary donations.

1.4 References

The project brief for CSE336, the Phase 1 assessment rubric, and the client requirements collected for this project. The design patterns referenced in this document are the six Phase 1 patterns: Factory, Decorator, Observer, Strategy, Singleton, and MVC.

2 Overall description

2.1 Product perspective

The system is a standalone desktop application. It replaces a manual process based on spreadsheets, paper forms, and handwritten attendance sheets. It does not depend on an external server or a network connection: the database is stored on the same computer, and the application starts without configuration.

Internally the product is organised in layers. The interface layer collects input and displays results, the controller layer receives user actions, the service layer applies the business rules,

and the repository layer stores and retrieves data. This organisation is described in the Phase 1 report and is treated here as a constraint on the implementation.

2.2 User classes

User class	Access	Main needs
Charity administrator	Full	Manage all records, reference data, and reports.
Event organiser	Events and attendance	Create events, register attendees, check attendance, send reminders.
Volunteer coordinator	Volunteers and assignments	Track availability, skills, hours, and certificates.
Donor or volunteer	Indirect	Receive receipts, statements, certificates, and notifications.

Phase 1 assumes a single operator with full access. Authentication and role enforcement are outside this phase.

2.3 Operating environment

- a desktop or laptop computer with a standard keyboard and display;
- a Java runtime and the packaged application;
- no network connection, external database server, or third-party account;
- English interface and Egyptian pound (EGP) as the currency.

2.4 Constraints and assumptions

- the project is an academic deliverable and must remain understandable to a student audience;
- payment processing and message delivery are simulated, because real providers require credentials and contracts;
- only the six Phase 1 patterns are required; later patterns belong to Phase 2;
- the data set is small (a single branch), so the system is not designed for concurrent multi-user access;
- the operator is assumed to be trained on the main screens.

3 Functional requirements

Each requirement has an identifier, a priority (Must or Should), and an acceptance condition that can be checked in the running application. Requirements marked Must are part of the Phase 1 demonstration.

3.1 Donations and donors

ID	Requirement	Priority	Acceptance
FR-D1	Create, edit, and archive donor records with name, contact details, and type.	Must	A donor can be added, edited, archived, and found again.
FR-D2	Record a donation as online, check, or in-kind.	Must	Each type saves the fields it needs and appears in the list.
FR-D3	Validate donation data: positive amount, check number for checks, item and quantity for in-kind.	Must	Invalid input is rejected with a clear message.
FR-D4	Filter donations by donor, type, and date range.	Must	The table shows only matching records.
FR-D5	Produce a receipt with optional tax and thank-you sections.	Must	The preview contains the selected sections.
FR-D6	Produce a yearly statement per donor from monetary donations.	Must	The total matches the donor's monetary donations for the year.
FR-D7	Process an online card or wallet payment through a simulated provider.	Should	A payment result is recorded and linked to the donation.

3.2 Volunteers

ID	Requirement	Priority	Acceptance
FR-V1	Create and edit volunteers with contact details and availability.	Must	The record is saved and the availability is visible.
FR-V2	Add and remove skills for a volunteer.	Must	The volunteer's skill list updates immediately.
FR-V3	Assign a volunteer to an open event only when available, skilled, and within capacity.	Must	Ineligible assignments are refused with the reason.
FR-V4	Record worked hours per volunteer and event.	Must	Hours are stored and included in the summary.
FR-V5	Show the total hours for a volunteer.	Must	The total equals the sum of recorded hours.
FR-V6	Generate a certificate when the volunteer reaches the required hours.	Must	A certificate is produced with the correct total.

3.3 Events and attendance

ID	Requirement	Priority	Acceptance
FR-E1	Create and edit events with title, date, place, capacity, and type.	Must	The event is saved and listed with its status.
FR-E2	Open and close registration for an event.	Must	Status changes are reflected and invalid transitions are refused.
FR-E3	Register an attendee and issue a ticket reference.	Must	The registration appears with a unique reference.
FR-E4	Check attendees in and produce an attendance report.	Must	Checked-in counts match the registrations.
FR-E5	Notify registered attendees when an event is updated, rescheduled, opened, or closed.	Must	A notification is recorded for each registered attendee.
FR-E6	Send a reminder for an upcoming event.	Should	A reminder is recorded for the event's attendees.

3.4 Beneficiaries and distribution

ID	Requirement	Priority	Acceptance
FR-B1	Create and edit beneficiaries with household size, need type, and status.	Must	The record is saved and the need type is selectable.
FR-B2	Record a distribution with item, quantity, date, optional donation link, and notes.	Must	The distribution is saved and refused for a closed beneficiary.
FR-B3	Show the distribution history for one beneficiary.	Must	The history lists exactly that beneficiary's distributions.
FR-B4	Distribute a quantity across eligible beneficiaries using a selected policy.	Should	The allocation follows the chosen policy and is recorded per beneficiary.
FR-B5	Record an impact score for a beneficiary.	Should	The score is saved and displayed.

3.5 Communication

ID	Requirement	Priority	Acceptance
FR-C1	Select email, SMS, or social media as the active channel at runtime.	Must	The selected channel is used by the next message.
FR-C2	Send a message to a recipient with a subject and body.	Must	The message is recorded and shown in the activity feed.
FR-C3	Keep a record of sent notifications for review.	Should	Recent notifications are visible in the interface.

3.6 Reference data (lookup tables)

ID	Requirement	Priority	Acceptance
FR-R1	Create, read, update, and delete event types.	Must	All four operations succeed and the event form reflects them.
FR-R2	Create, read, update, and delete skills.	Must	All four operations succeed and the volunteer form reflects them.
FR-R3	Reject duplicate names.	Must	A duplicate name is refused with a message.
FR-R4	Refuse deletion of a value that is still referenced.	Must	The value remains and the user is told why.

3.7 Reports and documents

ID	Requirement	Priority	Acceptance
FR-P1	Preview a receipt, yearly statement, attendance report, certificate, and distribution history.	Must	Each document opens with the correct data for the selection.

4 Non-functional requirements

ID	Category	Requirement	Priority
NFR-1	Usability	A new operator can complete the main tasks (record a donation, register a volunteer, create an event) after a short demonstration, without reading the code.	Must
NFR-2	Usability	Every screen uses consistent labels, and destructive actions ask for confirmation.	Must
NFR-3	Reliability	A failed database operation leaves no partial data; transactions are rolled back.	Must
NFR-4	Data integrity	Required fields are validated, and references between records are enforced.	Must
NFR-5	Performance	List and search operations return within one second for a single branch's data volume.	Should
NFR-6	Maintainability	The interface, control logic, business rules, and storage are separated, so a module can change without editing the others.	Must
NFR-7	Testability	Business rules can be tested without starting the graphical interface.	Should
NFR-8	Portability	The application runs on Windows, Linux, and macOS with a standard Java runtime.	Should
NFR-9	Capacity	The system stores the records of one branch and does not require concurrent multi-user access.	Must
NFR-10	Localisation	The interface is in English and amounts are shown in EGP.	Should

5 External interfaces and data

5.1 User interfaces

The product offers two interfaces over the same core:

- a **graphical interface** with a dashboard and one screen per module (donations, payments, donors, volunteers, events, beneficiaries, messages, and reference data);
- a **console interface** with a numbered menu, kept as a fallback and as evidence that the business rules do not depend on one presentation layer.

Both interfaces use the same terminology and the same confirmation behaviour for destructive actions.

5.2 Hardware and software interfaces

Interface	Requirement
Display	A standard desktop display; the graphical interface is designed for a window of at least 1024 by 700 pixels.
Input	Keyboard and mouse.
Runtime	A standard Java runtime on Windows, Linux, or macOS.
Storage	A local relational database file created automatically on first run.
Network	Not required for any Phase 1 function.

5.3 Communication interfaces

Email, SMS, and social media are represented by three interchangeable channels. In this version the channels are simulated: a message is validated, recorded, and displayed instead of being transmitted. The interface is designed so that a real provider can replace the simulation without changing the communication screen or the services that send messages.

5.4 Data requirements

Entity	Main data
Donor	Name, type, email, phone, status.
Donation	Donor, type, amount or item, date, reference, notes.
Payment	Donation, method, status, transaction reference.
Volunteer	Name, contact details, availability.
Skill	Name.
Volunteer skill	Volunteer and skill pair.
Volunteer hours	Volunteer, event, date, hours.
Event	Title, type, date, place, capacity, status.
Event type	Name.
Registration	Event, attendee name, email, ticket reference, check-in state.
Beneficiary	Name, phone, household size, need type, status, notes.
Need type	Name.
Distribution	Beneficiary, optional donation, item, quantity, date, notes.

The main relationships are: a donor has many donations; a donation has at most one payment; a volunteer has many skills and many hour records; an event has one type and many registrations; a beneficiary has one need type and many distributions; a distribution may be linked to one donation.

6 Traceability and scope limits

6.1 Client requirements to functional requirements

Client	Requirement	Functional requirements
CR1	Yearly tax statement per donor	FR-D6, FR-P1
CR2	Validate volunteer skill and availability before assignment	FR-V3
CR3	Certificate after the required cumulative hours	FR-V4, FR-V5, FR-V6
CR4	Notify attendees when an event is updated or rescheduled	FR-E5, FR-C2
CR5	Record aid distribution per beneficiary	FR-B2, FR-B3, FR-P1
CR6	Full CRUD for event types, and for skills	FR-R1, FR-R2, FR-R3, FR-R4
CR7	Email, SMS, and social channels, swappable at runtime	FR-C1, FR-C2

6.2 Out of scope for this version

The following items are deliberately excluded from Phase 1. They are listed so that the boundary of the deliverable is clear:

- user accounts, login, and role-based access control;
- real payment processing and real message delivery;
- multi-branch operation and concurrent multi-user access;
- mobile or web clients;
- the Phase 2 design patterns and the admin pages.

6.3 Acceptance of the phase

The phase is accepted when the Must requirements in Section 3 can be demonstrated from the running application, the reference-data module performs all four operations on event types and skills, and the six Phase 1 patterns are present in the class diagram and in the code.